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Filter circuit

Abstract

A filter circuit includes a resonant circuit, and the resonant circuit includes an inductance element and a capacitor. The capacitor includes an element stacked body, an exterior body, a first external electrode and a second external electrode, and a third external electrode. The element stacked body is formed as a stack of a plurality of capacitor elements. In a first capacitor element, a first end portion is electrically connected to the first external electrode. In a second capacitor element, a first end portion is electrically connected to the second external electrode. A third external electrode is electrically connected to a cathode part of the capacitor elements.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to a filter circuit, and more particularly to a filter circuit including a resonant circuit.

BACKGROUND

(2) A digital signal processor board disclosed in Unexamined Japanese Patent Publication No. 2006-352059 includes an LSI to which an element for clock operation is connected, a power input line that supplies power to the LSI, and a decoupling capacitor that is connected between the power input line and ground. As the decoupling capacitor, a surface-mounted solid electrolytic capacitor having an ESR of 25 mΩ (100 kHz) or lower and an ESL of 800 pH (500 MHz) or lower is used.

SUMMARY OF THE INVENTION

(3) A filter circuit according to one aspect of the present disclosure includes a resonant circuit. The resonant circuit includes an input end, an output end, an inductance element, and a capacitor. The inductance element electrically connects between the input end and the output end. The capacitor electrically connects between the inductance element and a conductor that is at a reference potential. The capacitor is an electrolytic capacitor. The capacitor includes an element stacked body, an exterior body, a first external electrode, a second external electrode, and a third external electrode. The element stacked body is a stack of a plurality of capacitor elements. The exterior body includes a sealing member covering the element stacked body. The first external electrode and the second external electrode are electrically connected to the inductance element. The third external electrode is electrically connected to the conductor that is at the reference potential. Each of the plurality of capacitor elements includes an anode body, a dielectric layer, and a cathode part. The anode body includes a porous part on a surface of the anode body. The dielectric layer is disposed on a surface of at least a part of the porous part. The cathode part covers at least a part of the dielectric layer. In a first end portion of the each of the plurality of capacitor elements, the anode body is exposed from the exterior body. In a second end portion of the each of the plurality of capacitor elements, the anode body is covered by the cathode part. The plurality of capacitor elements includes a first capacitor element and a second capacitor element. In the first capacitor element, the first end portion is electrically connected to the first external electrode. In the second capacitor element, the first end portion is electrically connected to the second external electrode. The third external electrode is electrically connected to the cathode part of the capacitor element.

(4) The present disclosure is advantageous in that the number of capacitors can be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a circuit diagram of an electric circuit including a filter circuit according to an exemplary embodiment of the present disclosure.
- (2) FIG. 2 is a plan view illustrating the filter circuit mounted on a substrate.
- (3) FIG. 3 is a perspective view of an electrolytic capacitor included in the filter circuit.
- (4) FIG. 4 is a perspective view of the electrolytic capacitor.
- (5) FIG. 5 is a schematic cross-sectional view of the electrolytic capacitor.
- (6) FIG. 6 is a schematic cross-sectional view of a capacitor element of the electrolytic capacitor.
- (7) FIG. 7 is a characteristic diagram of the filter circuit.
- (8) FIG. 8 is a perspective view of an electrolytic capacitor according to a first modification of the present disclosure.
- (9) FIG. 9 is a perspective view of an electrolytic capacitor according to a second modification of the present disclosure.
- (10) FIG. 10 is a perspective view of the electrolytic capacitor according to the second modification of the present disclosure.
- (11) FIG. 11 is a perspective view of an electrolytic capacitor according to another aspect of the second modification of the present disclosure.
- (12) FIG. 12 is a perspective view of an electrolytic capacitor according to a third modification of

the present disclosure.

(13) FIG. **13** is a perspective view of the electrolytic capacitor according to the third modification of the present disclosure.

(14) FIG. **14** is a perspective view of an electrolytic capacitor according to another aspect of the third modification of the present disclosure.

(15) FIG. **15** is a perspective view of an electrolytic capacitor according to a fourth modification of the present disclosure.

(16) FIG. **16** is a perspective view of the electrolytic capacitor according to the fourth modification of the present disclosure.

(17) FIG. **17** is a perspective view of an electrolytic capacitor according to another aspect of the fourth modification of the present disclosure.

(18) FIG. **18** is a circuit diagram of an electric circuit including a filter circuit according to a fifth modification of the present disclosure.

DESCRIPTION OF EMBODIMENT

(19) Before explaining the exemplary embodiment, a problem in a conventional technology will be briefly explained. Conventional electric circuits are sometimes provided with a filter circuit to reduce noise. However, conventionally, such a filter circuit has often been required to include many capacitors, too, to satisfy specification requirements of the filter circuit.

(20) In view of the problem described above, the present disclosure provides a filter circuit capable of reducing the number of capacitors.

(21) A filter circuit according to an exemplary embodiment will now be described with reference to drawings. The following exemplary embodiment is, however, merely illustrative of various embodiments of the present disclosure. The exemplary embodiment described below can be changed variously depending on factors such as a design, as long as the object of the present disclosure can be achieved. The drawings described in the following exemplary embodiment are schematic representations, and the ratio of the sizes and the thicknesses of each component in the drawings does not necessarily need to reflect the actual dimensional ratio.

Exemplary Embodiment

(22) (Overview)

(23) As illustrated in FIG. **1**, filter circuit **100** according to the present exemplary embodiment includes resonant circuit **101**. Resonant circuit **101** includes input end **T1**, output end **T2**, inductance element **42**, and capacitor **11**. Inductance element **42** electrically connects between input end **T1** and output end **T2**. Capacitor **11** electrically connects between inductance element **42** and conductor **W3** that is at a reference potential (see FIG. **2**). Capacitor **11** is an electrolytic capacitor.

(24) As illustrated in FIGS. **2** to **6**, capacitor **11** includes an element stacked body, exterior body **14**, first external electrode **21** and second external electrode **22**, and third external electrode **23** (**231** and **232**). The element stacked body is formed as a stack of a plurality of capacitor elements **10**. Exterior body **14** includes sealing member **140** covering the element stacked body. First external electrode **21** and second external electrode **22** are electrically connected to inductance element **42**. Third external electrode **23** is electrically connected to conductor **W3** that is at the reference potential. Each of the plurality of capacitor elements **10** includes anode body **3**, a dielectric layer, and cathode part **6**. Anode body **3** includes porous part **5** on a surface of anode body **3**. The dielectric layer is disposed on a surface of at least a part of porous part **5**. Cathode part **6** covers at least a part of the dielectric layer. In first end portion **1a** of the each of the plurality of capacitor elements **10**, anode body **3** is exposed from exterior body **14**. In second end portion **2a** of the each of the plurality of capacitor elements **10**, anode body **3** is covered by cathode part **6**. The plurality of capacitor elements **10** include first capacitor element **10a** and second capacitor element **10b**. In first capacitor element **10a**, first end portion **1a** is electrically connected to first external electrode **21**. In second capacitor element **10b**, first end portion **1a** is electrically connected to second external electrode **22**. Third external electrode **23** is electrically connected to cathode part **6** of

capacitor elements **10**.

(25) According to the present exemplary embodiment, currents flow in different directions in first capacitor element **10a** and second capacitor element **10b**, respectively, and these currents generate magnetic fields in different directions, respectively. Therefore, the magnetic flux generated in the element stacked body is reduced. In this manner, equivalent series inductance (ESL) of capacitor **11** is reduced.

(26) Further, since the ESL is reduced, the number of capacitors **11** can be reduced, compared with a configuration including a plurality of capacitors each having an ESL higher than that of capacitor **11** are connected in parallel.

(27) First capacitor element **10a** and second capacitor element **10b** are stacked alternately. As a result, the magnetic flux generated in the element stacked body can be reduced effectively. Therefore, it is possible to reduce the ESL effectively.

(28) First end portion **1a** of first capacitor element **10a** is exposed from first surface **14a** of exterior body **14**, and first end portion **1a** of second capacitor element **10b** is exposed from second surface **14b** that is opposite to first surface **14a** of exterior body **14**. As a result, an effect of first capacitor element **10a** and second capacitor element **10b** canceling out the magnetic fluxes is enhanced, so that the ESL of capacitor **11** can be reduced effectively.

(29) Further, at least a part of first external electrode **21** is provided on first surface **14a** of exterior body **14**, and at least a part of second external electrode **22** is provided on second surface **14b** that is opposite to first surface **14a** of exterior body **14**. At least a part of third external electrode **23** is provided on third surface **14c** different from first surface **14a** and second surface **14b** of exterior body **14**.

(30) In the following description related to capacitor **11**, the side having second surface **14b** is defined as “left” with respect to first surface **14a**, and the side having first surface **14a** is defined as “right” with respect to second surface **14b**. In addition, the side having fourth surface **14d** that is opposite to third surface **14c** is defined as an “upper” side with respect to third surface **14c**, and the side having third surface **14c** is defined as a “lower” side with respect to fourth surface **14d**.

Furthermore, the sides in the direction orthogonal to both upper and lower sides and left and right sides are defined as “front and rear” sides, respectively. As illustrated in FIG. 3, exterior body **14** has fifth surface **14e** (front surface) and sixth surface **14f** (rear surface). The height of capacitor **11** means the length of capacitor **11** in the up-down direction. However, these provisions are not intended to limit the directions in which capacitor **11** is used in any way.

(31) Capacitor **11** is mounted on substrate **B1** (see FIG. 2) in use, for example. Capacitor **11** is mounted on substrate **B1** in such a manner that the bottom surfaces of first external electrode **21**, second external electrode **22**, and third external electrode **23** are in contact with the surface of substrate **B1**. In this manner, first external electrode **21**, second external electrode **22**, and third external electrode **23** are electrically connected to substrate **B1**.

(32) In capacitor **11** according to the present exemplary embodiment, since at least a part of first external electrode **21** and at least a part of second external electrode **22** are provided on the side surfaces of exterior body **14**, it is possible to reduce the portions not contributing to the electrostatic capacitance inside capacitor **11**. Therefore, the capacitance of capacitor **11** can be increased. By increasing the capacitance of capacitor **11**, the number of capacitors **11** can be reduced compared with a configuration in which a plurality of capacitors each having a relatively small capacitance are connected in parallel to ensure the capacitance.

(33) (Details)

(34) (1) Circuit Configuration

(35) To begin with, filter circuit **100** and electric circuit **EC1** including filter circuit **100** according to the present exemplary embodiment will be described with reference to FIG. 1.

(36) Electric circuit **EC1** supplies power to integrated circuit **41**. Electric circuit **EC1** includes filter circuit **100**, DC/DC converter **43**, and conductors serving as a plurality of circuits **30**. The plurality

of circuits **30** include first circuit **31** and second circuit **32**.

(37) Integrated circuit **41** is, for example, a semiconductor integrated circuit (IC) or a large-scale integrated circuit (LSI). Integrated circuit **41** is included in a personal computer, a server computer, or a microcontroller, for example.

(38) A first end of first circuit **31** is electrically connected to power source **PS1**. In other words, electric circuit **EC1** is electrically connected to power source **PS1**. A second end of first circuit **31** is electrically connected to an input terminal of DC/DC converter **43**. Power source **PS1** is a DC power source. Power source **PS1** is a battery, for example. Power source **PS1** supplies DC power to DC/DC converter **43**.

(39) A first end of second circuit **32** is electrically connected to an output terminal of DC/DC converter **43**. A second end of second circuit **32** is electrically connected to integrated circuit **41**. In other words, electric circuit **EC1** is electrically connected to integrated circuit **41**. DC/DC converter **43** converts DC power input from power source **PS1** into DC power having a predetermined voltage, and outputs the DC power to integrated circuit **41**.

(40) Filter circuit **100** is provided on first circuit **31**. In other words, filter circuit **100** electrically connects between power source **PS1** and the input terminal of DC/DC converter **43**.

(41) Filter circuit **100** includes resonant circuit **101**. Filter circuit **100** according to the present exemplary embodiment constitute only of resonant circuit **101**.

(42) Resonant circuit **101** includes input end **T1**, output end **T2**, inductance element **42**, and capacitor **11**. Capacitor **11** includes two internal capacitors **111**, **112**. Two internal capacitors **111**, **112** are housed in one exterior body **14** (see FIG. 3) of capacitor **11**. A plurality of first capacitor elements **10a** (see FIG. 5), to be described later, are included in internal capacitor **111**, and a plurality of second capacitor elements **10b** (see FIG. 5) are included in internal capacitor **112**.

(43) Power source **PS1** is electrically connected to input end **T1**. The input terminal of DC/DC converter **43** is electrically connected to output end **T2**. Input end **T1** and output end **T2** are part of the conductor making up first circuit **31**. Input end **T1** and output end **T2** are connectors, and may be connected to other connectors provided to first circuit **31**.

(44) Inductance element **42** is a coil, for example. Inductance element **42** electrically connects between input end **T1** and output end **T2**. In other words, a first end of inductance element **42** is electrically connected to input end **T1**, and a second end of inductance element **42** is electrically connected to output end **T2**.

(45) An anode of internal capacitor **111** is electrically connected to the first end of inductance element **42** via first external electrode **21**. More particularly, the anode of internal capacitor **111** is electrically connected to a connection point between the first end of inductance element **42** and input end **T1**. A cathode of internal capacitor **111** is electrically connected to conductor **W3** (see FIG. 2) that is at the reference potential via third external electrode **231**. In the present exemplary embodiment, the reference potential is the ground potential.

(46) An anode of internal capacitor **112** is electrically connected to the second end of inductance element **42** via second external electrode **22**. More particularly, the anode of internal capacitor **112** is electrically connected to a connection point between the second end of inductance element **42** and output end **T2**. The cathode of internal capacitor **112** is electrically connected to conductor **W3** (see FIG. 2) that is at the reference potential via third external electrode **232**.

(47) In the manner described above, resonant circuit **101** forms π type filter. In other words, first external electrode **21** is electrically connected to the first end of inductance element **42**, and second external electrode **22** is electrically connected to the second end of inductance element **42**.

(48) Electric circuit **EC1** also includes capacitors **45**, **48**, in addition to capacitor **11**. Each of the capacitors **45**, **48** is an electrolytic capacitor. Electric circuit **EC1** also includes inductor **44**.

(49) Inductor **44** electrically connects between the output terminal of DC/DC converter **43** and capacitor **45**. Capacitor **45** electrically connects between second circuit **32** and conductor **W3** (ground) (see FIG. 2) that is at the reference potential. Inductor **44** and capacitor **45** together form a

low-pass filter.

(50) Capacitor **48** electrically connects between second circuit **32** and conductor **W3** (ground) at the reference potential. Capacitor **48** smooths the output voltage of DC/DC converter **43**.

(51) (2) Implementation of Capacitor and Inductance Element

(52) As illustrated in FIG. 2, capacitor **11** and inductance element **42** are mounted on the surface of substrate **B1**. Substrate **B1** has conductors **W1**, **W2**, and **W3**. Conductor **W1** includes input end **T1** (see FIG. 1) of resonant circuit **101**. Conductor **W2** includes output end **T2** (see FIG. 1) of resonant circuit **101**. The potential of conductor **W3** is the reference potential (ground potential). Conductor **W3** is provided between conductor **W1** and conductor **W2**.

(53) Capacitor **11** is mounted on substrate **B1** in such a manner that third surface **14c** that is the bottom surface (see FIG. 4) faces a surface of substrate **B1**. First external electrode **21** of capacitor **11** is electrically connected to conductor **W1**. Second external electrode **22** is electrically connected to conductor **W2**. Third external electrodes **231**, **232** are electrically connected to conductor **W3**.

(54) Since first external electrode **21** and second external electrode **22** are equivalent, first external electrode **21** may also be electrically connected to conductor **W2**, and second external electrode **22** may also be electrically connected to conductor **W1**.

(55) Inductance element **42** includes first electrode **421** serving as the first end and second electrode **422** serving as the second end. First electrode **421** is electrically connected to conductor **W1**. Second electrode **422** is electrically connected to conductor **W2**.

(56) (3) Characteristics of Capacitors and Resonant Circuit

(57) As mentioned above, capacitor **11** includes two internal capacitors **111**, **112**. The characteristics of capacitor **11** are defined as the characteristics of the parallel circuit of two internal capacitors **111**, **112**.

(58) Capacitor **11** has an equivalent series inductance (ESL) of 100 [pH] or lower, under the condition of a frequency of 50 [MHz] or higher and 1000 [MHz] or lower.

(59) Capacitor **11** has a capacitance of 10 [μ F] or more.

(60) As illustrated in FIG. 5, capacitor **11** has a stacked structure in which first capacitor element **10a** and second capacitor element **10b** are stacked, as a configuration for reducing the ESL. More specifically, capacitor **11** has a stacked structure in which first capacitor element **10a** and second capacitor element **10b** are stacked alternately.

(61) As a configuration for increasing the electrostatic capacitance, capacitor **11** has an end-face current collector structure in which at least a part of first external electrode **21** and at least a part of second external electrode **22** are provided on respective side surfaces of exterior body **14**.

(62) The solid line in FIG. 7 represents an attenuation characteristic of resonant circuit **101** including capacitor **11** and inductance element **42**. The broken line in FIG. 7 is an attenuation characteristic of a resonant circuit according to a comparative example. The resonant circuit according to the comparative example includes inductance element **42** and two-terminal conductive aluminum electrolytic capacitors (capacitance of 220 μ F) provided on the respective sides of inductance element **42**. In the present exemplary embodiment, by using capacitor **11** having a low ESL, the amount of attenuation can be increased across a wider band.

(63) (4) Structure of Capacitor

(64) A structure of capacitor **11** (hereinafter, also referred to as electrolytic capacitor **11**) according to the present exemplary embodiment will now be described with reference to FIGS. 5 and 6.

(65) [Electrolytic Capacitor]

(66) Electrolytic capacitor **11** according to the exemplary embodiment of the present disclosure includes an element stacked body that is a stack of a plurality of capacitor elements **10**, exterior body **14**, first external electrode **21**, second external electrode **22**, and third external electrode **23** (**231** and **232**). Exterior body **14** includes sealing member **140** covering the element stacked body. Exterior body **14** according to the present exemplary embodiment further includes substrate **17**. The bottom surface of substrate **17** serves as the bottom surface (third surface **14c**) of exterior body

14.

(67) Each of the plurality of capacitor elements **10** includes anode body **3** having a surface including porous part **5**, a dielectric layer provided on at least a part of a surface of porous part **5**, and cathode part **6** covering at least a part of the dielectric layer. Each of the plurality of capacitor elements **10** has first end portion **1a** where anode body **3** is exposed, and second end portion **2a** where anode body **3** is covered by cathode part **6**, and at least the end face of first end portion **1a** is exposed from exterior body **14**.

(68) In some of the plurality of capacitor elements **10**, first end portions **1a** face first surface **14a** of exterior body **14**, and in the others, first end portions **1a** face second surface **14b** that is different from first surface **14a** of exterior body **14**. Among such capacitor elements **10**, one having first end portion **1a** facing first surface **14a** of exterior body **14** will be referred to as first capacitor element **10a**, and one having first end portion **1a** facing second surface **14b** that is different from first surface **14a** of exterior body **14** will be referred to as second capacitor element **10b**. First end portion **1a** of first capacitor element **10a** is electrically connected to first external electrode **21**. First end portion **1a** of second capacitor element **10b** is electrically connected to second external electrode **22**.

(69) With this configuration, currents flow through first capacitor element **10a** and second capacitor element **10b** in different directions with respect to each other. As a result, the magnetic fields are generated by the currents in different directions, so that the magnetic flux generated in the element stacked body decreases. Therefore, the ESL is reduced. Preferably, first surface **14a** and second surface **14b** may be surfaces of exterior body **14** facing each other. Furthermore, the magnetic flux generated in the element stacked body can be reduced effectively by stacking first capacitor element **10a** and second capacitor element **10b** alternately. Therefore, it is possible to reduce the ESL effectively.

(70) The number of first capacitor elements **10a** and the number of second capacitor elements **10b** may be the same. When the number of first capacitor elements **10a** is the same as the number of second capacitor elements **10b**, the magnetic field generated by the current flowing through first capacitor element **10a** and the magnetic field generated by the current flowing through second capacitor element **10b** cancel out each other without any over or short, so that the magnetic flux generated in the element stacked body is reduced. In this manner, the ESL can be reduced easily.

(71) Furthermore, the element stacked body and the external electrode may be electrically connected by electrically connecting the end face of first end portion **1a** of each capacitor element **10** exposed from exterior body **14** to the external electrode (first external electrode **21** or second external electrode **22**). The end face of first end portion **1a** and the external electrode may be electrically connected by, for example, using an external electrode provided on first surface **14a** or second surface **14b**, or by electrically connecting an intermediate electrode (corresponding to anode electrode layer **16** described later) provided on first surface **14a** or second surface **14b**, to the external electrode. With such a configuration, it is not necessary to interpose another member for connecting first end portion **1a** to the external electrode (first external electrode **21** or second external electrode **22**) inside exterior body **14**, so that the capacity of electrolytic capacitor **11** can be increased easily. In addition, in the current path from a part of anode body **3** not having cathode part **6** (anode lead-out part) to first external electrode **21** or second external electrode **22**, the length of the path of current flowing in parallel with a stacked surface of the element stacked body is substantially equal to that of the anode lead-out part, and therefore, can be easily shortened. Therefore, the ESL resultant of the path of the current flowing in parallel with the stack surface of the element stacked body can be further reduced.

(72) Third external electrode **23** (**231** and **232**) is electrically connected to cathode part **6** of capacitor element **10**. For example, third external electrode **23** (**231** and **232**) is electrically connected to cathode part **6** on the outermost layer (that is, the lowermost layer or the uppermost layer) of the element stacked body. With this, a cathode terminal may be provided on the bottom

surface of electrolytic capacitor **11**. An anode terminal may also be provided on the bottom surface of electrolytic capacitor **11** by extending first external electrode **21** or second external electrode **22** onto the bottom surface of electrolytic capacitor **11**. In this configuration, the current flowing through the extended portion of first external electrode **21** or second external electrode **22** flows in the opposite direction of the current flowing through the anode lead-out portions. Therefore, the magnetic field generated by the current flowing through the anode lead-out portions is canceled by the magnetic field generated by the current flowing through the extended portion of first external electrode **21** or second external electrode **22**, so that the ESL of electrolytic capacitor **11** is further reduced. As a result, because the extended portion can shorten the distance by which the cathode terminal is separated from the first and/or second external electrodes, the ESL is improved. With these synergistic effects, the ESL is reduced significantly.

(73) First end portion **1a** may be electrically connected to first external electrode **21** or second external electrode **22** via contact layer **15**. Contact layer **15** may be selectively formed on the end faces of first end portions **1a** of the plurality of respective capacitor elements **10**, for example. Contact layer **15** may connect first end portions **1a** of the plurality of respective capacitor elements **10** to an intermediate electrode (anode electrode layer **16**) or an external electrode provided in a manner covering first surface **14a** or second surface **14b**. By interposing contact layer **15**, electrical connection between first end portions **1a** and the external electrode can be established reliably. Therefore, reliability of electrolytic capacitor **11** can be improved.

(74) First external electrode **21** and second external electrode **22** may face each other in the longitudinal direction of anode body **3**, or may face each other in the short-hand direction. For example, first external electrode **21** and second external electrode **22** may be disposed on respective ends in the short-hand direction of one surface (e.g., the bottom surface) of exterior body **14**, or may be disposed on respective ends of the longitudinal direction. In terms of reducing the ESL, first external electrode **21** and second external electrode **22** may face each other in the short-hand direction of anode body **3**. By contrast, by disposing first external electrode **21** and second external electrode **22** in a manner facing each other in the longitudinal direction of anode body **3**, it is easier to increase the distance by which first external electrode **21** or second external electrode **22** extends onto the bottom surface of electrolytic capacitor **11**, to control the distance by which the cathode terminal is separated from the anode terminal, and to control the ESL to a desired level.

(75) FIG. 5 is a cross-sectional view schematically illustrating a structure of electrolytic capacitor **11** according to the exemplary embodiment of the present disclosure. FIG. 6 is a cross-sectional view illustrating a structure of capacitor element **10** included in electrolytic capacitor **11** in FIG. 5. However, electrolytic capacitor **11** according to the present disclosure is not limited thereto.

(76) As illustrated in FIGS. 5 and 6, electrolytic capacitor **11** includes a plurality of capacitor elements **10** (**10a**, **10b**). Each capacitor element **10** includes anode body **3** and cathode part **6**. Anode body **3** is, for example, a foil (anode foil). Anode body **3** has a surface including porous part **5**, and porous part **5** has a surface at least a part of which is provided with a dielectric layer (not illustrated). Cathode part **6** covers at least a part of the dielectric layer.

(77) In capacitor element **10**, one end (first end portion) **1a** is not covered by cathode part **6**, and anode body **3** is exposed, whereas anode body **3** on the other end (second end portion) **2a** is covered by cathode part **6**. Hereinafter, the part of anode body **3** not covered by cathode part **6** will be referred to as first part **1**, and the part of anode body **3** covered by cathode part **6** will be referred to as second part **2**. An end of first part **1** corresponds to first end portion **1a**, and an end of second part **2** corresponds to second end portion **2a**. The dielectric layer is formed on a surface of porous part **5** that is provided at least on second part **2**. Note that, first part **1** of anode body **3** is also referred to as an anode lead-out portion. Second part **2** of anode body **3** is also referred to a cathode forming portion.

(78) More specifically, second part **2** includes core **4** and porous part (porous body) **5** formed by roughening (e.g., etching) the surface of core **4**. First part **1** may, however, have or not have porous

part 5 on the surface. The dielectric layer is provided along the surface of porous part 5. At least a part of the dielectric layer covers and is provided on inner wall surfaces of the pores of porous part 5.

(79) Cathode part 6 includes solid electrolyte layer 7 covering at least a part of the dielectric layer, and a cathode lead-out layer covering at least a part of solid electrolyte layer 7. A surface of the dielectric layer has irregularities corresponding to the shape of the surface of anode body 3. Solid electrolyte layer 7 may have a shape by which such irregularities of the dielectric layer are filled. The cathode lead-out layer includes, for example, carbon layer 8 covering at least a part of solid electrolyte layer 7, and silver paste layer 9 covering carbon layer 8.

(80) Second part 2 corresponds to a part of anode body 3 provided with solid electrolyte layer 7 with the dielectric layer (porous part 5) therebetween, and first part 1 corresponds to a part of anode body 3 on which no solid electrolyte layer 7 is provided with the dielectric layer (porous part 5) therebetween.

(81) Insulating separation layer (or insulating member) 12 may be provided in a manner covering the surface of anode body 3, the part being a part of the area of anode body 3 not facing the cathode part 6 but sharing a border with cathode part 6. This separation layer restricts the contact between cathode part 6 and the exposed part (first part 1) of anode body 3. Separation layer 12 is, for example, an insulating resin layer.

(82) In the example illustrated in FIG. 5, four capacitor elements 10 (10a, 10b) are stacked in such a manner that cathode parts 6 (second parts 2) thereof are placed on top of one another. However, there are two types of capacitor elements 10 having first part 1 of anode body 3 facing different directions. In FIG. 5, each first capacitor element 10a has anode body 3 with first part 1 facing one direction (rightwards in the drawing) with respect to second part 2. By contrast, each second capacitor element 10b has anode body 3 with first part 1 facing the direction opposite to the direction in which first part 1 of first capacitor element 10a faces (leftwards in the drawing), with respect to second part 2. First capacitor elements 10a and second capacitor elements 10b are stacked alternately, to form an element stacked body. In the plurality of capacitor elements 10 (10a, 10b), cathode parts 6 that are adjacent to each other in the stacked direction are electrically connected to each other via conductive adhesive layers 13. Adhesive layer 13 is formed using a conductive adhesive, for example. Adhesive layer 13 contains, for example, silver.

(83) Electrolytic capacitor 11 includes the above-described element stacked body in which a plurality of capacitor elements 10 (10a, 10b) are stacked, exterior body 14 including sealing member 140 for covering the element stacked body, first external electrode 21, second external electrode 22, and third external electrode 23. In the element stacked body, the end face of first end portion 1a is exposed from exterior body 14.

(84) Exterior body 14 has a substantially rectangular cuboid outer shape, and electrolytic capacitor 11 also has a substantially cuboid outer shape. Exterior body 14 has first surface 14a and second surface 14b opposite to first surface 14a. In the element stacked body, first end portion 1a of each first capacitor element 10a faces first surface 14a (that is, first end portion 1a is closer to first surface 14a than the second end portion 2a), and first end portion 1a of each second capacitor element 10b faces second surface 14b (that is, first end portion 1a is closer to second surface 14b than the second end portion 2a).

(85) In electrolytic capacitor 11, each of the plurality of first end portions 1a (first parts 1) exposed from exterior body 14 is electrically connected to first external electrode 21 that extends along first surface 14a or second external electrode 22 that extends along second surface 14b. With such a configuration, it is not necessary to bundle the plurality of first parts 1 to form an anode of electrolytic capacitor 11, nor to ensure a length for bundling the plurality of first parts 1. In this manner, first parts 1 occupy a smaller proportion of anode body 3, so that a capacitance can be improved compared with a configuration in which the plurality of first parts 1 are bundled. In addition, the ESL contributed by first parts 1 is reduced. In addition, because the distance by which

third external electrode **23** is separated from first external electrode **21** and/or second external electrode **22** can be shortened, the ESL is improved.

(86) In electrolytic capacitor **11**, each of end faces of the plurality of first end portions **1a** exposed from exterior body **14** is covered by contact layer **15**. Anode electrode layer **16** covers contact layers **15**, and first surface **14a** and second surface **14b** of exterior body **14**. First external electrode **21** and second external electrode **22** cover anode electrode layer **16**, so that the plurality of first end portions **1a** (first parts **1**) are electrically connected to first external electrode **21** or second external electrode **22** thereby. Specifically, anode electrode layer **16** covering first surface **14a** of exterior body **14** is interposed between contact layer **15** and first external electrode **21**, and anode electrode layer **16** covering second surface **14b** of exterior body **14** is interposed between contact layer **15** and second external electrode **22**.

(87) In the example in FIG. 5, the element stacked body is supported on substrate **17**. Substrate **17** is, for example, a layered substrate the upper surface and the lower surface of which have conductive wiring patterns, and the wiring pattern on the upper surface and the wiring pattern on the lower surface are electrically connected by a through hole. The wiring pattern on the upper surface is electrically connected to cathode part **6** of capacitor element **10** in the lowermost layer of the stack, and the wiring pattern on the lower surface (third surface **14c**) is electrically connected to third external electrode **23** (**231** and **232**). Therefore, third external electrode **23** is electrically connected to cathode part **6** of each capacitor element **10** of the element stacked body via substrate **17**. In such a configuration, the number, shape, and arrangement of the third external electrodes **23** may be set in any way, depending on the wiring pattern provided on the lower surface. Third external electrode **23** is provided on substrate **17** by plating, for example, so that substrate **17** including third external electrode **23** may be handled as one member.

(88) At least a part of third external electrode **23** is exposed on the bottom surface of electrolytic capacitor **11**. The exposed portion on the bottom surface of third external electrode **23** serves as a cathode terminal of electrolytic capacitor **11**. In the example in FIG. 5, two third external electrodes **23** are provided separated from each other, and third external electrodes **23** are exposed in a plurality of respective areas.

(89) A part of first external electrode **21** is bent along the bottom surface of exterior body **14** and exposed on the bottom surface of electrolytic capacitor **11**. Similarly, a part of second external electrode **22** is bent along the bottom surface of exterior body **14**, in a manner facing the bent part of first external electrode **21**, and is exposed on the bottom surface of electrolytic capacitor **11**. These exposed portions of first external electrode **21** and second external electrode **22** on the bottom surface serve as an anode terminal of electrolytic capacitor **11**. In other words, in the present exemplary embodiment, electrolytic capacitor **11** has two anode terminals that are separated from each other. A cathode terminal may be provided in a manner interposed between the two anode terminals that are separated from each other.

(90) The ESL of electrolytic capacitor **11** depends on distance **L1** by which first external electrode **21** is separated from third external electrode **23** on the bottom surface, and distance **L2** by which second external electrode **22** is separated from third external electrode **23** on the bottom surface. When separation distances **L1** and **L2** are shorter, it is more likely for the ESL to be smaller. In order to reduce the ESL, a plurality of third external electrodes **23** may be disposed on the bottom surface. In such a configuration, one of the plurality of third external electrodes **23** (third external electrode **231**) may be disposed closer to first external electrode **21**, and the other one of the plurality of third external electrodes **23** (third external electrode **232**) may be disposed closer to second external electrode **22**. In this manner, the ESL can be reduced effectively. Separation distances **L1** and **L2** may be, for example, 0.4 mm to 1.1 mm. Note that “including a plurality of third external electrodes **23**” means that third external electrodes **23** are exposed in a plurality of separated areas, and is not limited to a configuration in which the plurality of third external electrodes **23** are provided separated from each other. Two or more of the plurality of third external

electrodes **23** may be continuous and electrically connected inside exterior body **14**. The plurality of third external electrodes **23** may be provided on different surfaces of exterior body **14**, e.g., one provided on the top surface and the other provided on the bottom surface.

(91) As illustrated in FIG. **4**, in a view in the normal direction of third surface **14c** (bottom surface), distance **L1** between edge **2310** of third external electrode **23** facing first external electrode **21** and first external electrode **21** is preferably shorter than distance **L3** between line segment **S1** that is orthogonal to the direction in which first surface **14a** and second surface **14b** are arranged (left-right direction) and that divides exterior body **14** into halves, and edge **2310** of third external electrode **23**. When a plurality of third external electrodes **23** are provided as in the present exemplary embodiment, edge **2310** of third external electrode **23** facing first external electrode **21** means an edge of third external electrode **231** having an edge facing first external electrode **21**, among the plurality of third external electrodes **23**.

(92) Furthermore, in a view in the normal direction of third surface **14c** (bottom surface), distance **L2** between edge **2320** of third external electrode **23** facing second external electrode **22** and second external electrode **22** is preferably shorter than distance **L4** between line segment **S1** that is orthogonal to the direction in which first surface **14a** and second surface **14b** are arranged (left-right direction) and that divides exterior body **14** into two halves, and edge **2320** of third external electrode **23**. When a plurality of third external electrodes **23** are provided as in the present exemplary embodiment, edge **2320** of third external electrode **23** facing second external electrode **22** means an edge of third external electrode **232** having an edge facing second external electrode **22**, among the plurality of third external electrodes **23**.

(93) In electrolytic capacitor **11**, the direction of the current flowing through first capacitor element **10a** is opposite to the direction of the current flowing through second capacitor element **10b**. Thus, a magnetic field generated by a current flowing through first capacitor element **10a** and a magnetic field generated by a current flowing through second capacitor element **10b** cancel out each other, and a magnetic flux generated in electrolytic capacitor **11** decreases. As a result, the ESL is reduced.

(94) At the same time, in parts of first part **1** and second part **2** where capacitor elements **10** do not overlap each other (in FIG. **5**, a portion not covered by the cathode lead-out layer), the effect of canceling out the magnetic fields does not occur, but in electrolytic capacitor **11** according to the present exemplary embodiment, it is easy to reduce the length of first part **1**. Thus, contributions of these parts to the ESL is reduced. Furthermore, since first external electrode **21** and second external electrode **22** extend along the bottom surface of exterior body **14**, contributions of these parts to the ESL is reduced. With these effects, the ESL of electrolytic capacitor **11** can be improved significantly.

(95) These elements of electrolytic capacitor **11** according to the exemplary embodiment will now be described more in detail.

(96) (Anode Body **3**)

(97) Anode body **3** can contain a valve metal, an alloy containing a valve metal, and a compound containing a valve metal (such as an intermetallic compound), for example. These materials can be used singly, or two or more of these materials may be used in combination. As the valve metal, aluminum, tantalum, niobium, or titanium may be used, for example. Anode body **3** may be a foil of a valve metal, an alloy containing a valve metal, or a compound containing a valve metal, or may be a porous sintered body of a valve metal, an alloy containing a valve metal, or a compound containing a valve metal.

(98) When a metal foil is used for anode body **3**, porous part **5** is usually formed at least on a surface of second part **2** of anode foil to increase a surface area. Second part **2** includes core **4** and porous part **5** formed on a surface of the core **4**. Porous part **5** may be formed by roughening, e.g., etching the surface of at least second part **2** of the anode foil. It is also possible to dispose a predetermined masking member on the surface of first part **1** before roughening the surface by

etching, for example. It is also possible to roughen the entire surface of the anode foil by etching. In the former case, the resultant anode foil has no porous part **5** on the surface of first part **1** and has porous part **5** on the surface of second part **2**. In the latter case, porous part **5** is formed not only on the surface of second part **2** but also on the surface of first part **1**. As the etching, any known method may be used, and an example thereof includes electrolytic etching. The masking member is not limited to any particular member, but an insulator such as resin is preferable. The masking member may be a conductor containing a conductive material, although the masking member is removed before solid electrolyte layer **7** is formed.

(99) When the entire surface of the anode foil is roughened, porous part **5** is formed on the surface of first part **1**. Therefore, adhesion between porous part **5** and sealing member **140** may become insufficient, and air (specifically, oxygen and moisture) may get into electrolytic capacitor **11** through the part where porous part **5** is in contact with sealing member **140**. In order to suppress insufficiency in the adhesion, porous first part **1** may be compressed in advance, so that the pores of porous part **5** are crushed. In this manner, it is possible to suppress the entry of the air into electrolytic capacitor **11** from first end portion **1a** exposed from sealing member **140** through porous part **5**, and to suppress a reduction in the reliability of electrolytic capacitor **11** due to the entry of the air.

(100) (Dielectric Layer)

(101) The dielectric layer is formed, for example, by anodizing the valve metal at least on the surface of second part **2** of anode body **3**, by applying chemical conversion treatment thereto. The dielectric layer contains an oxide of the valve metal. For example, when aluminum is used as the valve metal, the dielectric layer contains aluminum oxide. The dielectric layer is formed at least on the surface of second part **2** having porous part **5** (including the inner wall surfaces of the pores of porous part **5**). The method of forming the dielectric layer is not limited to this, and any method may be used as long as an insulating layer functioning as a dielectric layer is formed on the surface of second part **2**. The dielectric layer may also be formed on the surface of first part **1** (e.g., on porous part **5** of the surface of first part **1**).

(102) (Cathode Part **6**)

(103) Cathode part **6** includes solid electrolyte layer **7** covering at least a part of the dielectric layer, and a cathode lead-out layer covering at least a part of solid electrolyte layer **7**.

(104) (Solid Electrolyte Layer **7**)

(105) Solid electrolyte layer **7** contains, for example, a conductive polymer. Examples of the conductive polymer include polypyrrole, polythiophene, polyaniline, and derivatives thereof. Solid electrolyte layer **7** can be formed on the dielectric layer by chemical polymerization and/or electrolytic polymerization of a raw material monomer, for example. Alternatively, the dielectric layer may also be formed by applying a solution in which the conductive polymer is dissolved or a dispersion liquid in which the conductive polymer is dispersed to the dielectric layer. Solid electrolyte layer **7** may contain a manganese compound.

(106) (Cathode Lead-Out Layer)

(107) The cathode lead-out layer includes, for example, a carbon layer **8** and silver paste layer **9**. Carbon layer **8** is only required to be conductive, and may be made of a conductive carbon material such as graphite, for example. Carbon layer **8** is formed by, for example, applying a carbon paste to at least a part of the surface of solid electrolyte layer **7**. For silver paste layer **9**, for example, a composition containing silver powder and a binder resin (e.g., epoxy resin) may be used. Silver paste layer **9** is formed by, for example, applying a silver paste onto the surface of carbon layer **8**. The configuration of the cathode lead-out layer is not limited thereto, and may be any configuration having a current collector function.

(108) (Separation Layer **12**)

(109) Insulating separation layer **12** may be provided to electrically separate first part **1** from cathode part **6**. Separation layer **12** may be provided near cathode part **6**, in a manner covering at

least a part of the surface of first part **1**. Preferably, separation layer **12** is in close adhesion with first part **1** and sealing member **140**. In this manner, entry of the air into electrolytic capacitor **11** can be suppressed. Separation layer **12** may be disposed on first part **1** with the dielectric layer therebetween.

(110) Separation layer **12** contains, for example, a resin, and resins to be explained later as examples of sealing member **140** may be used. The dielectric layer provided on porous part **5** of first part **1** may be compressed and densified to provide insulation.

(111) Separation layer **12** in close adhesion with first part **1** can be achieved by, for example, bonding a sheet-like insulating member (e.g. resin tape) to first part **1**. When an anode foil having porous part **5** on the surface is used, porous part **5** of first part **1** may be compressed and flattened before bringing the insulating member into close adhesion with first part **1**. The sheet-like insulating member preferably has an adhesive layer on a surface that is to be bonded to first part **1**.

(112) The insulating member in close adhesion with first part **1** may be achieved by applying liquid resin to first part **1**, or impregnating first part **1** therewith. In the method using the liquid resin, the insulating member is formed in such a manner that the irregularities on the surface of porous part **5** of first part **1** are filled thereby. Because liquid resin easily gets into the recesses on the surface of porous part **5**, inside of the recesses can be easily filled with the insulating member. As the liquid resin, a curable resin composition examples of which will be described in a fourth step below may be used.

(113) (Sealing Member **140**)

(114) Sealing member **140** of exterior body **14** preferably contains, for example, a cured product of a curable resin composition, and may contain a thermoplastic resin or a composition containing a thermoplastic resin.

(115) Sealing member **140** may be formed using, for example, a molding technique such as injection molding. Sealing member **140** may be formed, for example, by using a predetermined mold and filling a predetermined portion of the mold with a curable resin composition or a thermoplastic resin (composition) in such a manner that capacitor element **10** is covered thereby.

(116) The curable resin composition may contain a filler, a curing agent, a polymerization initiator, and/or a catalyst, in addition to the curable resin. Examples of the curable resin include thermosetting resins. The curing agent, the polymerization initiator, the catalyst, and the like are selected as appropriate, depending on the type of the curable resin.

(117) As the curable resin composition and the thermoplastic resin (composition), those examples of which will be described in a third step below may be used.

(118) From the viewpoint of adhesion between separation layer **12** and sealing member **140**, the insulating member and sealing member **140** preferably both contain resin. Sealing member **140** can adhere closely with an insulating member containing a resin, compared with first part **1** containing a valve metal, or the dielectric layer containing an oxide of the valve metal.

(119) More preferably, separation layer **12** and sealing member **140** contain the same resin. In this case, adhesion between separation layer **12** and sealing member **140** is further improved, so that the entry of the air into electrolytic capacitor **11** is further suppressed. An example of the same resin contained in separation layer **12** and sealing member **140** includes an epoxy resin.

(120) From the viewpoint of enhancing the strength and the like of sealing member **140**, preferably, sealing member **140** contains a filler. By contrast, it is preferable for separation layer **12** to contain a filler having a particle diameter smaller than that of sealing member **140**, and more preferable for the separation layer **12** not to contain any filler. When separation layer **12** is formed by impregnating first part **1** with a liquid resin, the liquid resin preferably contains a filler having a particle size smaller than that of sealing member **140**, and more preferably, the liquid resin does not contain any filler. With such a liquid resin, the recesses on the surface of porous part **5** of first part **1** can be deeply impregnated with the resin, so that separation layer **12** can be formed easily. In addition, liquid resin enables thin separation layer **12** to be formed so as to allow the plurality of

capacitor elements **10** to be stacked.

(121) (Contact Layer **15**)

(122) Contact layer **15** may be formed in a manner covering the end face of first end portion **1a** of anode body **3**. Preferably, contact layer **15** may be formed in a manner covering only the surface of first end portion **1a** that is exposed from sealing member **140**, and covering the surface of sealing member **140** (and separation layer **12**) that is a resin material as little as possible.

(123) Contact layer **15** may contain a metal having a lower ionization tendency than the metal from which anode body **3** is made. For example, when anode body **3** is made of aluminum (Al) foil, contact layer **15** may be made of a material containing Zn, Ni, Sn, Cu, or Ag, for example. Such a material inhibits formation of a strong oxide film on the surface of contact layer **15**, so that the electrical connection can be established more reliably, compared with the configuration in which the portion of anode body **3** exposed on first end portion **1a** is directly connected to the external electrode.

(124) An alloy layer may be formed on the interface between contact layer **15** and anode body **3**. For example, when anode body **3** is an aluminum (Al) foil, Cu, Zn, or Ag, having atomic spacing near that of Al, can form an alloy layer on the interface, as a result of intermetallic bonding with Al. In this manner, the strength of bonding with anode body **3** can be further enhanced. Contact layer **15** may be made from a single element metal of the element mentioned above or an alloy such as bronze or brass, or may be formed as a stack of a plurality of metal layers of different single elements (e.g., a layered structure of a Cu layer and an Ag layer).

(125) When contact layer **15** is to be formed, it is preferable for sealing member **140** not to contain a filler, or if sealing member **140** contains a filler, it is preferable for the Young's modulus of the filler to be lower than the Young's modulus of contact layer **15**. In this manner, formation of contact layer **15** on the surface of sealing member **140** is suppressed, and contact layer **15** can be formed selectively on the end face of first end portion **1a**.

(126) Contact layer **15** may be formed by cold spraying, thermal spraying, plating, or vapor deposition, for example. With cold spraying, for example, metal particles in the solid state are brought into collision with the surface (first surface **14a** and/or second surface **14b**) of the sealing member **140** where the surface of first end portion **1a** is exposed, to cause the metal particles to be fixed to the surface by plastic deformation, so that contact layer **15** containing the metal of the metal particles is formed on the end face of first end portion **1a**. At this time, if the Young's modulus of the metal particles is higher than the Young's modulus of a constituent of sealing member **140** (e.g., the filler), plastic deformations of the metal particles having collided with the surface of sealing member **140** on the surface of sealing member **140** are suppressed, so that the metal particles become less fixed to the surface of sealing member **140**. At least a part of energy resultant of the collision is used to break sealing member **140**, to scrape off a part of the resin. As a result, it is possible to form contact layer **15** selectively on the end face of first end portion **1a** of anode body **3**, and to roughen the surface of sealing member **140** (first surface **14a** and/or second surface **14b**) at the same time.

(127) (Anode Electrode Layer **16**)

(128) Anode electrode layer **16** may be interposed between contact layer **15** and the external electrode (first external electrode **21** or second external electrode **22**). Anode electrode layer **16** may cover first surface **14a** or second surface **14b** of exterior body **14**, and may be electrically connected to first end portion(s) **1a** of capacitor element(s) **10** via contact layer **15**, as necessary.

(129) Anode electrode layer **16** may include a layer of conductive resin having conductive particles mixed. The conductive resin layer can be formed by applying a conductive paste containing conductive particles and a resin material onto first surface **14a** or second surface **14b** of exterior body **14**, and drying the conductive paste. A resin material is suitable for being bonded to the material of exterior body **14** and anode body **3** (contact layer **15**), and capable of enhancing bonding strength by chemical bonding (e.g., hydrogen bonding). As the conductive particles, for

example, metal particles such as those of silver or copper, or particles of a conductive inorganic material such as those of carbon may be used.

(130) Anode electrode layer **16** may be a metal layer. In that case, anode electrode layer **16** may be formed by using electrolytic plating, electroless plating, sputtering, vacuum vapor deposition, chemical vapor deposition (CVD), cold spraying, or thermal spraying.

(131) Anode electrode layer **16** may cover apart of a surface (e.g., atop or bottom surface) orthogonal to first surface **14a** and second surface **14b** of exterior body **14**.

(132) Roughness Ra of the surface of exterior body **14** covered by anode electrode layer **16** may be 5 micrometers or more. With such setting, the contact area between anode electrode layer **16** and exterior body **14** is increased, and the adhesion between anode electrode layer **16** and exterior body **14** is improved by the anchoring effect, so that the reliability can be further enhanced.

(133) (External Electrode)

(134) First to third external electrodes **21** to **23** are preferably metal layers. These metal layers are, for example, plated layers. The metal layers contain, for example, at least one selected from the group consisting of nickel (Ni), copper (Cu), zinc (Zn), tin (Sn), silver (Ag), and gold (Au). To form first to third external electrodes **21** to **23**, for example, a film formation technique such as an electrolytic plating, electroless plating, sputtering, vacuum vapor deposition, chemical vapor deposition (CVD), cold spraying, or thermal spraying may be used.

(135) Each of first to third external electrodes **21** to **23** may have a layered structure of a Ni layer and a tin layer, for example. Each of first to third external electrodes **21** to **23** may be any metal at least the outer surface of which has excellent wettability with solder. Examples of such a metal include Sn, Au, Ag, and Pd.

(136) Each of first external electrode **21** and second external electrode **22** may be formed by bonding a Cu cap, having a Sn film formed thereon in advance, to anode electrode layer **16**.

(137) First external electrode **21** and second external electrode **22** together form an anode terminal of electrolytic capacitor **11**. When electrolytic capacitor **11** is mounted on substrate **17**, both first external electrode **21** and second external electrode **22** need to be bonded to an electrode on substrate **17**. However, first external electrode **21** and second external electrode **22** may be electrically connected to each other via a surface of exterior body **14** other than first surface **14a** and second surface **14b**. In such a configuration, only one of first external electrode **21** and second external electrode **22** may be connected to the electrode on substrate **17** at the time of mounting electrolytic capacitor **11** on substrate **17**.

(138) [Method of Manufacturing Electrolytic Capacitor **11**]

(139) The electrolytic capacitor **11** according to the exemplary embodiment of the present disclosure can be manufactured by, for example, a manufacturing method including a first step of preparing anode body **3**; a second step of obtaining a plurality of capacitor elements **10**; a third step of obtaining an element stacked body that is a stack of the plurality of capacitor elements **10**; a fourth step of covering the element stacked body with sealing member **140**; a fifth step of forming an end face of first part **1** and exposing the end face from sealing member **140**; and a sixth step of electrically connecting the end face of first part **1** to an external electrode. The manufacturing method may further include a step of disposing separation layer **12** (insulator member) on a part of anode body **3** (separation layer disposing step). These steps included in the method of manufacturing electrolytic capacitor **11** will now be described.

(140) (First Step)

(141) In the first step, anode body **3** having a surface with a dielectric layer is prepared. More specifically, anode body **3** including first part **1** including one end and second part **2** including the other end opposite to one end and having a dielectric layer formed at least on a surface of second part **2** is prepared. For example, the first step includes a step of forming porous part **5** on a surface of anode body **3**, and a step of forming a dielectric layer on a surface of porous part **5**. More specifically, anode body **3** used in the first step has first part **1** including an end to be removed (the

one end) and second part **2** including second end portion **2a** (the other end). Preferably, porous part **5** is provided at least on the surface of second part **2**.

(142) Porous part **5** on the surface of anode body **3** may be formed in any way as long as irregularities can be formed on the surface of anode body **3**, e.g., by roughening the surface of anode foil by etching (e.g., electrolytic etching).

(143) The dielectric layer may be formed by applying chemical conversion treatment to anode body **3**. The chemical conversion treatment may be applied by, for example, immersing anode body **3** in a chemical conversion solution so that the surface of anode body **3** is impregnated with the chemical conversion solution, and applying a voltage between anode body **3** as an anode and a cathode immersed in the chemical conversion solution. When the surface of anode body **3** has porous part **5**, the dielectric layer is formed on the irregular surface of porous part **5**.

(144) (Separation Layer Disposing Step)

(145) When electrolytic capacitor **11** including separation layer **12** (insulating member) is to be manufactured, a step of disposing separation layer **12** (insulating member) may be performed, after the first step and before the second step. In this step, an insulating member is disposed on a part of anode body **3**. More specifically, in this step, an insulating member is disposed on first part **1** of anode body **3**, with a dielectric layer interposed therebetween. The insulating member is disposed so as to isolate first part **1** from cathode part **6** that is to be formed in a subsequent step.

(146) In the separation layer disposing step, a sheet-like insulating member (e.g., resin tape) may be bonded to a part (e.g., first part **1**) of anode body **3**. Even when anode body **3** having a surface with porous part **5** is used, by compressing and flattening the irregularities on the surface of first part **1**, it is possible to bring the insulating member into close adhesion with first part **1**. The sheet-like insulating member preferably has an adhesive layer on a surface that is to be bonded to first part **1**.

(147) In addition to the above, in the separation layer disposing step, an insulating member may be formed by coating or impregnating a part (e.g., first part **1**) of anode body **3** with a liquid resin. For example, the liquid resin is applied or anode body **3** is impregnated with liquid resin, and then the liquid resin is cured. With this, the insulating member can be brought into close adhesion with first part **1**, easily. As the liquid resin, a curable resin composition examples of which will be explained in the fourth step (of forming sealing member **140**), or a resin solution obtained by dissolving a resin in a solvent may be used, for example.

(148) When the surface of anode body **3** has porous part **5**, it is preferable to apply liquid resin to a part of the surface of porous part **5** of anode body **3** (e.g., the surface of first part **1**), or to impregnate such a part with liquid resin. In this manner, an insulating member filling the irregularities on the surface of porous part **5** of first part **1** can be formed easily. Because liquid resin easily gets into the recesses on the surface of porous part **5**, inside of the recesses can be easily filled with the insulating member. In this manner, the insulating member protects porous part **5** on the surface of anode body **3**, and suppresses collapse of porous part **5** of anode body **3** in the fourth step in which anode body **3** is partially removed with sealing member **140**. Since the insulating member is in close adhesion with the surface of porous part **5** of anode body **3**, the insulating member is less likely to peel off from the surface of porous part **5** of anode body **3** in the fourth step in which anode body **3** is partially removed with sealing member **140**.

(149) (Second Step)

(150) In the second step, cathode part **6** is formed on anode body **3** to obtain capacitor element **10**. When the insulating member is to be provided in the sixth step, cathode part **6** is formed in a portion of anode body **3** where the insulating member is not disposed in the second step, to obtain capacitor element **10**. More specifically, in the second step, at least a part of the dielectric layer on the surface of second part **2** of anode body **3** is covered by cathode part **6**.

(151) The step of forming cathode part **6** includes, for example, a step of forming a solid electrolyte covering at least a part of the dielectric layer, and a step of forming a cathode lead-out layer

covering at least a part of solid electrolyte layer **7**.

(152) Solid electrolyte layer **7** can be formed on the dielectric layer by chemical polymerization and/or electrolytic polymerization of a raw material monomer, for example. Solid electrolyte layer **7** may be formed by attaching a treatment liquid containing a conductive polymer to the dielectric layer, and then drying the treatment liquid. The treatment liquid may further contain other components such as a dopant. As the conductive polymer, poly (3,4-ethylenedioxythiophene) (PEDOT) is used, for example. As the dopant, for example, polystyrene sulfonic acid (PSS) is used. The treatment liquid is a dispersion liquid or a solution of the conductive polymer. Examples of the dispersion medium (solvent) include water, an organic solvent, and a mixture thereof.

(153) The cathode lead-out layer can be formed by, for example, sequentially layering carbon layer **8** and silver paste layer **9** on the solid electrolyte layer.

(154) (Third Step)

(155) In the third step, a plurality of capacitor elements **10** are stacked to obtain an element stacked body. In this step, for example, cathode parts **6** of the plurality of capacitor elements **10** are placed on top of each other, with a conductive adhesive interposed therebetween, in such a manner that first parts **1** of adjacent capacitor elements **10** face opposite directions alternately, to obtain an element stacked body.

(156) The element stacked body is then placed on a layered substrate (substrate **17**) the upper surface and the lower surface of which have wiring patterns, with a conductive adhesive interposed between the element stacked body and the substrate. Third external electrode **23** is formed in advance on the layered substrate on the side opposite to the side on which the element stacked body is placed. By placing the element stacked body, third external electrode **23** is electrically connected to cathode parts **6** of capacitor elements **10** included in the element stacked body, via the wiring patterns formed on the layered substrate and the through holes connecting the wiring patterns on the upper surface to the wiring patterns on the lower surface of the layered substrate.

(157) Alternatively, for example, the element stacked body and third external electrode **23** may be electrically connected by bonding plate-like third external electrode **23** having a predetermined shape to the surface of cathode part **6** exposed in the lowermost layer or the uppermost layer of the element stacked body, via a conductive paste or the like.

(158) Third external electrode **23** may be formed using electrolytic plating, electroless plating, physical vapor deposition, chemical vapor deposition, cold spraying, and/or thermal spraying.

(159) (Fourth Step)

(160) In the fourth step, the element stacked body is covered by sealing member **140**. At this time, third external electrode **23** is not entirely covered by sealing member **140**, and at least a part of third external electrode **23** remains exposed. Sealing member **140** may be formed by injection molding, for example. Sealing member **140** may be formed, for example, by using a predetermined mold and filling a predetermined portion with a curable resin composition or a thermoplastic resin (composition) in a manner covering the element stacked body.

(161) The curable resin composition may contain a filler, a curing agent, a polymerization initiator, and/or a catalyst, in addition to the curable resin. Examples of the curable resin include an epoxy resin, a phenol resin, a urea resin, polyimide, polyamideimide, polyurethane, diallyl phthalate, and unsaturated polyester. Examples of the thermoplastic resin include polyphenylene sulfide (PPS) and polybutylene terephthalate (PBT). It is also possible to use a thermoplastic resin composition containing a thermoplastic resin and a filler.

(162) As the filler, for example, insulating particles and/or fibers are preferable. Examples of the insulating material included in the filler include insulating compounds (e.g., oxides) such as silica and alumina, glass, and mineral materials (e.g., talc, mica, or clay). Sealing member **140** may contain one type of these fillers, or may contain two or more types of these fillers in combination.

(163) (Fifth Step)

(164) In the fifth step, after the fourth step, the end faces of the first parts **1** are formed and exposed

from the sealing member **140**. More specifically, at least a part of anode body **3** is removed together with sealing member **140** on both end sides of the element stacked body, and at least first end portion **1a** (specifically, the end face of first end portion **1a**) of anode body **3** is exposed from sealing member **140** on both of first surface **14a** and second surface **14b**. Examples of the method for exposing first end portion **1a** from sealing member **140** include a method of covering capacitor element **10** with sealing member **140**, and then polishing a surface of sealing member **140** or cutting off a part of sealing member **140** in such a manner that first end portion **1a** is exposed from sealing member **140**. A part of first part **1** may be cut off with a part of sealing member **140**. With this approach, first end portion **1a** having a surface having neither porous part **5** nor a natural oxide film can be exposed more easily from sealing member **140**, and a highly reliable connection with low resistance can be achieved between first part **1** and the external electrode. As a method for cutting sealing member **140**, dicing is preferable. As a result, the exposed end face of first end portion **1a** of first part **1** appears on the cut surface. Since the element stacked body includes two types of capacitor elements **10** with first parts **1** facing different directions, at the time of cutting off the parts of first parts **1** with a part of sealing member **140**, it is necessary to cut first parts **1** at two locations. One of the two cut faces becomes first surface **14a**, and the other becomes second surface **14b**.

(165) In the fifth step, anode body **3** and the insulating member may be partially removed with sealing member **140**, on both end sides of the element stacked body, to expose the end face of first end portion **1a** and the end face of the insulating member from sealing member **140**. In this case, an end face of anode body **3** becomes exposed in flush with an end face of the insulating member from sealing member **140**. Consequently, an end face of anode body **3** and end face of the insulating member can be easily exposed from sealing member **140** in flush with the surface of sealing member **140**.

(166) With the fifth step, it is possible to expose the end face of anode body **3** (first end portion **1a**) with no natural oxide film from sealing member **140**, and a highly reliable connection with low resistance between anode body **3** (more specifically, first part **1**) and the external electrode can be achieved.

(167) (Sixth Step)

(168) In the sixth step, the end face (first end portion **1a**) of each anode body **3** exposed from exterior body **14** is electrically connected to the external electrode. In this step, for example, first external electrode **21** is provided in a manner covering first surface **14a** of exterior body **14**, second external electrode **22** is provided in a manner covering second surface **14b**, and these external electrodes are electrically connected to the end face of first end portion **1a**. The electrical connection between the end face of first end portions **1a** and the external electrodes may be formed by joining, for example, or by electrolytic plating, electroless plating, physical vapor deposition, chemical vapor deposition, cold spraying, and/or thermal spraying.

(169) Before providing first external electrode **21** and second external electrode **22**, a step of forming contact layer **15** on a surface that is an end face of first end portion **1a** and/or a step of forming anode electrode layer **16** covering first surface **14a** or second surface **14b** of exterior body **14** may be performed. When anode electrode layer **16** is formed, first external electrode **21** and second external electrode **22** are formed in a manner covering anode electrode layer **16**.

(170) (Step of Forming Contact Layer **15**)

(171) Contact layer **15** may be formed by using methods such as cold spraying, thermal spraying, plating, or vapor deposition. Contact layer **15** may be formed in a manner selectively covering end face of first end portion **1a** and covering first surface **14a** and second surface **14b** of exterior body **14** as little as possible.

(172) When cold spraying is used, contact layer **15** is formed by causing metal particles to collide with the end face of each first end portion **1a** at high speed. The metal particles may be particles of a metal having a lower ionization tendency than that of the metal of anode body **3**. For example,

when anode body **3** is an Al foil, the metal particles may be Cu particles. Cu particles colliding with the end face of first end portion **1a** at high speed may break through the natural oxide film (Al oxide film) on the end face, and a metal bond between Al and Cu may be formed. As a result, a layer of an alloy of Al and Cu may be formed on the interface between contact layer **15** and first end portion **1a**. The surface of contact layer **15**, by contrast, is constituted by a layer of Cu that is a non-valve metal. Because the ionization tendency of Cu is lower than that of Al, the surface of contact layer **15** is less likely to oxidize. Therefore, it is possible to achieve a more reliable electrical connection between contact layer **15** and the external electrodes (or anode electrode layer **16**).

(173) Cold spraying is a technique in which metal particles having a size of several μm to several tens of μm are accelerated to a subsonic or supersonic speed using compressed gas such as air, nitrogen, or helium, and are caused to collide with a base material that is in a solid phase state, to form a metal film thereby. As to the mechanism how the metal particles stick by cold spraying, although there is somewhat unknown part, it is generally considered that the energy of colliding metal particles causes the metal particles or the metal substrate to go through a plastic deformation, and the newly formed surface become exposed and activated on the metal surface.

(174) With cold spraying, the metal particles may collide with first surface **14a** and second surface **14b** of exterior body **14**, and the end face of separation layer **12** (insulating member) that are made of a non-metallic material. When the metal particles collide with a base material made of a resin base material, it is considered that bonding of the metal particles to the resin base material take place due to mechanical bonding, i.e., mainly resultant of plastically deformed metal particles fitting into the irregularities on the surface of the resin base material. Therefore, in order for a metal film to be formed on the surface of a resin base material, the following conditions need to be satisfied: (ia) the resin base material has sufficient hardness so that the energy of collision is efficiently spent in causing plastic deformation of the metal particles; (iia) a metal material and working conditions are selected such that the metal particles are more likely to go through plastic deformation; and (iiia) the resin base material is less likely to collapse by receiving the energy of collision.

(175) By contrast, for the metal particles not to be fixed to the resin base material, following basic conditions need to be met: (ib) the resin base material is made elastic so that the colliding energy is not turned into an energy for causing a plastic deformation; (iib) a metal material and processing conditions are selected from a range allowing contact layer **15** to be formed on the end face of first end portion **1a** but being less likely to cause a plastic deformation of the metal particles; and (iiib) the strength of the resin base material is reduced so that the base material breaks with an impact at a level lower than that at which a plastic deformation occurs.

(176) In general, the plastic deformation of metal particles at the time of collision tends to be promoted when the Young's modulus of the metal particles is lower than the Young's modulus of the member (e.g., fillers) included in the resin base material, and the plastic deformation of the metal particles at the time of collision tends to be inhibited when the Young's modulus of the metal particles is higher. In the latter case, the collision energy of the metal particles fractures the resin base material brittlely, and the surface of the resin base material is scraped off thereby.

(177) Therefore, by making the Young's modulus of the metal particles (which may also be referred to as contact layer **15**) higher than the Young's modulus of the filler contained in the resin base material, it is possible to create a condition in which the metal particles hardly become fixed to the resin base material. In this manner, it is possible to inhibit formation of contact layer **15** on first surface **14a** and second surface **14b** of exterior body **14** and the end face of separation layer **12** (insulating member), and to form contact layer **15** selectively on the end face of first end portion **1a**. In addition, by causing the metal particles to collide with first surface **14a** and second surface **14b** of exterior body **14**, it is possible to achieve the effect of roughening first surface **14a** and second surface **14b**.

(178) For example, when silica having a Young's modulus of 94 GPa is filled as sealing member **140** of exterior body **14**, Cu particles and Ni particles may be used as metal particles having a Young's modulus higher than that of silica, and capable of easily being bonded with Al. However, the condition how the metal particles are fixed also varies depending on factors such as the shape, the size, and the temperature of the metal particles, and the size and filling rate of the silica filled in the resin material. Therefore, the present invention is not limited thereto.

(179) (Step of Forming Anode Electrode Layer **16**)

(180) Anode electrode layer **16** may be formed in a manner covering an end face of first end portion **1a** or contact layer **15**, and covering first surface **14a** and second surface **14b** of exterior body **14**. When separation layer **12** is provided, anode electrode layer **16** may be formed in a manner covering the end face of separation layer **12** (insulating member).

(181) Anode electrode layer **16** may be formed by applying a conductive paste containing conductive particles and a resin material. Specifically, a conductive paste (e.g., silver paste) is applied to each end surface by dipping, or transferring, printing, or dispensing the paste on the end face, for example, and then cured at a high temperature, so that anode electrode layer **16** is formed thereby.

(182) Anode electrode layer **16** that is a metal layer may also be formed by using a method such as electrolytic plating, electroless plating, sputtering, vacuum vapor deposition, chemical vapor deposition (CVD), cold spraying, or thermal spraying.

(183) (First Modification)

(184) A first modification related to the external electrodes will now be explained with reference to FIG. **8**. Configurations that are the same as those in the exemplary embodiment are given the same reference signs, and descriptions thereof will be omitted as appropriate.

(185) Capacitor **11** according to the first modification is different from that of the exemplary embodiment in including four third external electrodes **23**. Main parts of two of third external electrodes **23** are provided on fifth surface **14e** (front surface) of exterior body **14**. Main parts of the remaining two of third external electrodes **232** are provided on sixth surface **14f** (rear surface) of exterior body **14**.

(186) Each third external electrode **23** has a length in the up-down direction. The upper end of each of the third external electrodes **23** is bent toward the center of exterior body **14**. The lower end of each of the third external electrodes **23** is bent toward the center of exterior body **14**. As a result, a part of each third external electrode **23** is provided on third surface **14c** (bottom surface). Another part of each third external electrode **23** is provided on fourth surface **14d** (top surface).

(187) (Second Modification)

(188) A second modification related to the external electrodes will now be explained with reference to FIGS. **9** and **10**. Configurations that are the same as those in the exemplary embodiment are given the same reference signs, and descriptions thereof will be omitted as appropriate.

(189) Capacitor **11** according to this second modification is different from that of the exemplary embodiment in that two third external electrodes **23** (**231** and **232**) have different configurations. A main part of third external electrode **231** is provided on fifth surface **14e** (front surface) of exterior body **14**. A main part of third external electrode **232** is provided on sixth surface **14f** (rear surface) of exterior body **14**.

(190) These two third external electrodes **23** have lengths in the up-down direction. The upper end of each of the third external electrodes **23** is bent toward the center of exterior body **14**. The lower end of each of the third external electrodes **23** is bent toward the center of exterior body **14**. As a result, a part of each third external electrode **23** is provided on third surface **14c** (bottom surface). Another part of each third external electrode **23** is provided on fourth surface **14d** (top surface).

(191) As another modification of this second modification, as illustrated in FIG. **11**, two third external electrodes **23** in FIG. **10** may be connected on third surface **14c** (bottom surface) to form one third external electrode **23**.

(192) (Third Modification)

(193) A third modification related to the external electrodes will now be explained with reference to FIGS. 12 and 13. Configurations that are the same as those in the exemplary embodiment are given the same reference signs, and descriptions thereof will be omitted as appropriate.

(194) Capacitor 11 according to this third modification is different from that of the exemplary embodiment in that first external electrode 21 and second external electrode 22 have different configurations. Each of first external electrode 21 and second external electrode 22 has a U-shape with its connected portion facing downwards. Each of first external electrode 21 and second external electrode 22 covers a part of third surface 14c (bottom surface) of exterior body 14, a part of fifth surface 14e (front surface), and a part of sixth surface 14f (rear surface). First external electrode 21 and second external electrode 22 also cover parts of fourth surface 14d (top surface).

(195) The shape of third external electrodes 23 is similar to that of the exemplary embodiment. In other words, each third external electrode 23 has a U-shape with its connected portion facing downwards, and covers a part of third surface 14c (bottom surface), a part of fifth surface 14e (front surface), and a part of sixth surface 14f (rear surface). Each third external electrode 23 also cover a part of fourth surface 14d (top surface).

(196) As a still further modification of this third modification, as illustrated in FIG. 14, four third external electrodes 23 may be provided instead of two third external electrodes 23 in FIG. 13. Since the configurations of four third external electrodes 23 are the same as those illustrated in FIG. 8, descriptions thereof will be omitted.

(197) (Fourth Modification)

(198) A fourth modification related to the external electrodes will now be explained with reference to FIGS. 15 and 16. Configurations that are the same as those in the exemplary embodiment are given the same reference signs, and descriptions thereof will be omitted as appropriate.

(199) Capacitor 11 according to this fourth modification is different from that of the exemplary embodiment in that first external electrode 21, second external electrode 22, and two third external electrodes 23 have different configurations.

(200) Since the configurations of first external electrode 21 and second external electrode 22 are the same as those illustrated in FIG. 12, descriptions thereof will be omitted. Since the configurations of two third external electrodes 23 are the same as those illustrated in FIG. 10, descriptions thereof will be omitted.

(201) As still another modification of this fourth modification, as illustrated in FIG. 17, two third external electrodes 23 in FIG. 16 may be connected on third surface 14c (bottom surface) to form one third external electrode 23.

(202) (Fifth Modification)

(203) Fifth modification related to the circuit configuration of filter circuit 100 using capacitor 11 will now be explained with reference to FIG. 18.

(204) Resonant circuit 101 forms an L-type filter. In other words, the first end of inductance element 42 is electrically connected to input end T1. First external electrode 21 of capacitor 11 is electrically connected to the second end of inductance element 42, and second external electrode 22 is also electrically connected to the second end of inductance element 42 as well. More specifically, first external electrode 21 and second external electrode 22 are electrically connected to connection points between the second end of inductance element 42 and output end T2.

(205) In this manner, by using capacitor 11 having a relatively low ESL to provide the L-type filter, the amount by which signals attenuate can be increased across a relatively wide band.

(206) Note that resonant circuit 101 may further include a second inductance element different from inductance element 42 (first inductance element). A first end of the second inductance element may be electrically connected to second external electrode 22, and a second end of the second inductance element may be electrically connected to output end T2. In other words, resonant circuit 101 may form a T-type filter.

(207) (Other Modifications of Exemplary Embodiment)

(208) Other modifications of the exemplary embodiment will be listed below. The following modifications may also be implemented in combinations, as appropriate. In addition, the following modifications may be implemented in a manner combined with the modifications described above, as appropriate.

(209) In the exemplary embodiment, the number of capacitors **11** forming the alternating stacked structure and the end face collector structure is one. However, filter circuit **100** may include a plurality of capacitors **11**. In addition, a parallel circuit of the plurality of capacitors **11** may electrically connects between the circuit **30** and conductor **W3** that is at the reference potential.

(210) As power source **PS1**, it is also possible to use an AC power source and an AC/DC converter that converts AC power input from the AC power source into DC power, and that outputs the DC power.

(211) The number of sets of first capacitor element **10a** and second capacitor element **10b** is not limited to two, and may be one or three or more. In addition, first capacitor element **10a** and second capacitor element **10b** are not limited to those stacked alternately, and it is possible to include alternate stacking of a first stacked structure including a plurality of first capacitor elements **10a** and a second stacked structure including a plurality of second capacitor elements **10b**.

(212) Exterior body **14** may include casing for housing sealing member **140**.

SUMMARY

(213) Based on the exemplary embodiment and the like described above, the following aspects are disclosed.

(214) Filter circuit (**100**) according to first aspect includes resonant circuit (**101**). Resonant circuit (**101**) includes input end (**T1**), output end (**T2**), inductance element (**42**), and capacitor (**11**).

Inductance element (**42**) electrically connects between input end (**T1**) and output end (**T2**).

Capacitor (**11**) electrically connects between inductance element (**42**) and conductor (**W3**) that is at a reference potential. Capacitor (**11**) is an electrolytic capacitor. Capacitor (**11**) includes an element stacked body, exterior body (**14**), first external electrode (**21**) and second external electrode (**22**), and third external electrode (**23**). The element stacked body is formed as a stack of a plurality of capacitor elements (**10**). Exterior body (**14**) includes sealing member (**140**) covering the element stacked body. First external electrode (**21**) and second external electrode (**22**) are electrically connected to inductance element (**42**). Third external electrode (**23**) is electrically connected to conductor (**W3**) that is at the reference potential. Each of the plurality of capacitor elements (**10**) includes anode body (**3**), a dielectric layer, and cathode part (**6**). Anode body (**3**) includes a surface with porous part (**5**). The dielectric layer is disposed on a surface of at least a part of the porous part (**5**). Cathode part (**6**) covers at least a part of the dielectric layer. In first end portion (**1a**) of the each of the plurality of capacitor elements (**10**), anode body (**3**) is exposed from exterior body (**14**). In second end portion (**2a**) of the each of the plurality of capacitor elements (**10**), anode body (**3**) is covered by cathode part (**6**). The plurality of capacitor elements (**10**) include first capacitor element (**10a**) and second capacitor element (**10b**). In first capacitor element (**10a**), first end portion (**1a**) is electrically connected to first external electrode (**21**). In second capacitor element (**10b**), first end portion (**1a**) is electrically connected to second external electrode (**22**). Third external electrode (**23**) is electrically connected to cathode part (**6**) of capacitor element (**10**).

(215) With the configuration described above, the ESL of capacitor (**11**) can be reduced. Therefore, because the ESL is reduced, the number of capacitors can be reduced, compared with a configuration including a plurality of capacitors having an ESL higher than that of the capacitors (**11**) are connected in parallel.

(216) In filter circuit (**100**) according to a second aspect, in the first aspect, at least a part of first external electrode (**21**) is provided on first surface (**14a**) of exterior body (**14**). At least a part of second external electrode (**22**) is provided on second surface (**14b**) that is opposite to first surface (**14a**) of exterior body (**14**). At least a part of third external electrode (**23**) is provided on third

surface (14c) that is different from first surface (14a) and second surface (14b) of exterior body (14).

(217) With the configuration described above, the capacitance of capacitor (11) can be increased. Therefore, the number of capacitors can be reduced, compared with a configuration in which a plurality of capacitors each having a relatively small capacitance are connected in parallel to ensure the capacitance.

(218) In filter circuit (100) according to a third aspect, in the second aspect, in viewing in the normal direction of third surface (14c), distance (L1) between first external electrode (21) and edge (2310) of third external electrode (23) facing first external electrode (21) is shorter than distance (L2) between line segment (S1) and edge (2310) of third external electrode (23). Line segment (S1) is orthogonal to a direction in which first surface (14a) and second surface (14b) are arranged, and divides the exterior body (14) into two halves.

(219) With the configuration described above, because distance (L1) between third external electrode (23) and first external electrode (21) is relatively short, the ESL of capacitor (11) is improved.

(220) In filter circuit (100) according to a fourth aspect, in any one of the first to third aspects, an input terminal of DC/DC converter (43) is electrically connected to output end (T2).

(221) With the configuration described above, the noise of the input power of DC/DC converter (43) can be reduced.

(222) In filter circuit (100) according to a fifth aspect, in any one of the first to the fourth aspects, first external electrode (21) is electrically connected to the first end of inductance element (42). Second external electrode (22) is electrically connected to the second end of inductance element (42).

(223) With the configuration described above, since n type filter can be formed with one capacitor (11), the number of capacitors can be reduced, compared with a configuration in which two capacitors are used to form n type filter.

(224) In filter circuit (100) according to a sixth aspect, in any one of the first to the fifth aspects, an equivalent series inductance of capacitor (11) is 100 [pH] or lower under a condition of a frequency ranging from 50 [MHz] to 1000 [MHz], inclusive.

(225) With the configuration described above, the number of capacitors can be reduced.

(226) In filter circuit (100) according to a seventh aspect, in any one of the first to sixth aspects, a capacitance of capacitor (11) is 10 [μ F] or more.

(227) With the configuration described above, the number of capacitors can be reduced.

(228) The configurations other than the first aspect are not essential to filter circuit (100), and may be omitted as appropriate.

(229) Further, a configuration according to the following eighth aspect may be implemented without requiring the configurations of the first aspect. Filter circuit (100) according to an eighth aspect includes resonant circuit (101). Resonant circuit (101) includes input end (T1), output end (T2), inductance element (42), and capacitor (11). Inductance element (42) electrically connects between input end (T1) and output end (T2). Capacitor (11) electrically connects between inductance element (42) and conductor (W3) that is at a reference potential. Capacitor (11) has an equivalent series inductance of 100 [pH] or lower, under a condition of a frequency of 50 [MHz] or higher and 1000 [MHz] or lower.

(230) With the configuration described above, the number of capacitors can be reduced.

Claims

1. A filter circuit comprising a resonant circuit, the resonant circuit including: an input end; an output end; an inductance element electrically connecting between the input end and the output end; and a capacitor electrically connecting between the inductance element and a conductor that is

at a reference potential, wherein: the capacitor is an electrolytic capacitor, and the capacitor includes: an element stacked body that is a stack of a plurality of capacitor elements, an exterior body that includes a sealing member covering the element stacked body, a first external electrode and a second external electrode that are electrically connected to the inductance element, and a third external electrode electrically connected to the conductor that is at the reference potential, each of the plurality of capacitor elements includes: an anode body including a porous part on a surface of the anode body, a dielectric layer disposed on a surface of at least a part of the porous part, and a cathode part covering at least a part of the dielectric layer, in a first end portion of the each of the plurality of capacitor elements, the anode body is exposed from the exterior body, and in a second end portion of the each of the plurality of capacitor elements, the anode body is covered by the cathode part, the plurality of capacitor elements include a first capacitor element with the first end portion electrically connected to the first external electrode, and a second capacitor element with the first end portion electrically connected to the second external electrode, and the third external electrode is electrically connected to the cathode part of the capacitor element.

2. The filter circuit according to claim 1, wherein: at least a part of the first external electrode is provided on a first surface of the exterior body, at least a part of the second external electrode is provided on a second surface of the exterior body, the second surface being opposite to the first surface, and at least a part of the third external electrode is provided on a third surface of the exterior body, the third surface being different from the first surface and the second surface.

3. The filter circuit according to claim 2, wherein, in viewing in a normal direction of the third surface, a distance between the first external electrode and an edge of the third external electrode, the edge being an edge facing the first external electrode, is shorter than a distance between a line segment and the edge of the third external electrode, the line segment being orthogonal to a direction in which the first surface and the second surface are arranged and dividing the exterior body into two halves.

4. The filter circuit according to claim 1, wherein an input terminal of a DC/DC converter is electrically connected to the output end.

5. The filter circuit according to claim 1, wherein: the first external electrode is electrically connected to a first end of the inductance element, and the second external electrode is electrically connected to a second end of the inductance element.

6. The filter circuit according to claim 1, wherein an equivalent series inductance of the capacitor is 100 pH or lower, under a condition of a frequency ranging from 50 MHz to 1000 MHz, inclusive.

7. The filter circuit according to claim 1, wherein a capacitance of the capacitor is 10 μ F or more.
